# Companion XLI: The Epoch of Reionization in the Relaxation-Driven Cyclic Cosmology

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**Abstract**

The Epoch of Reionization (EoR) marks the transition from a neutral to an ionized Universe and represents one of the most sensitive probes of early structure formation. In the standard cosmological model, reionization is driven by the formation of the first stars and galaxies, whose ultraviolet radiation ionizes the intergalactic medium. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the formation history of early galaxies, the propagation of ionization fronts, and the topology of reionization. This Companion develops a continuous description of reionization in RDCC, deriving how the relaxation scale influences the timing, morphology, and observational signatures of the EoR. The resulting framework provides a natural interpretation of the early appearance of massive galaxies and the extended ionization structures observed in deep surveys.

## 1 Introduction

The Epoch of Reionization is a pivotal phase in cosmic history. After recombination, the Universe remained neutral for several hundred million years, until the first stars and galaxies formed and emitted ultraviolet photons capable of ionizing hydrogen. The resulting ionized bubbles expanded, merged, and eventually filled the Universe.

In the standard cosmological model, the timing and morphology of reionization are determined by the formation rate of early galaxies and the clumpiness of the intergalactic medium. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the growth of small-scale modes and enhances the coherence of large-scale modes. This modifies the formation history of early galaxies and the propagation of ionization fronts, imprinting the relaxation scale $k_{\rm rel}$ onto the structure of the EoR.

## 2 Early Galaxy Formation in RDCC

The formation of the first galaxies is governed by the collapse of dark matter halos. In RDCC, the gravitational potential takes the form

$$\Phi_{\rm RDCC}(k,a) = \Phi(k,a)\left[1 - \chi_\Phi \left(\frac{k}{k_{\rm rel}}\right)^\alpha\right]. \quad (1)$$

Large-scale modes, which dominate the collapse of massive halos, retain their coherence. As a result, massive halos form earlier in RDCC than in the standard cosmological model. Small-scale modes, which would normally produce numerous low-mass halos, are suppressed, delaying the formation of dwarf galaxies.

The halo mass function at high redshift becomes

$$n_{\mathrm{RDCC}}(M,z) = n(M,z)\left[1 - \chi_n\left(\frac{k(M)}{k_{\mathrm{rel}}}\right)^{\alpha}\right]. \tag{2}$$

This modification naturally explains the early appearance of massive galaxies observed by JWST.

# 3 Ionization Fronts and the RDCC Radiation Field

Ionization fronts propagate through the intergalactic medium according to

$$\frac{dR_{\mathrm{I}}}{dt} = \frac{\dot{N}_\gamma}{4\pi R_{\mathrm{I}}^2 n_{\mathrm{H}}} - \alpha_{\mathrm{B}} n_{\mathrm{H}} R_{\mathrm{I}}, \tag{3}$$

where $\dot{N}_\gamma$ is the ionizing photon production rate.

In RDCC, the altered formation history of early galaxies modifies $\dot{N}_\gamma$. Massive galaxies, which form earlier, produce strong ionizing radiation fields, while the suppressed formation of dwarf galaxies reduces the contribution from low-mass halos.

The resulting ionization fronts are larger, smoother, and more coherent than in the standard cosmological model. Their expansion reflects the coherence of the gravitational potential on scales near $k_{\mathrm{rel}}$.

# 4 Topology of Reionization in RDCC

The topology of reionization describes the spatial distribution of ionized and neutral regions. In RDCC, the suppression of small-scale modes reduces the fragmentation of ionized bubbles, while the enhanced coherence of large-scale modes promotes the formation of extended ionized regions.

The ionized volume fraction evolves as

$$Q_{\mathrm{HII,RDCC}}(z) = Q_{\mathrm{HII}}(z)\left[1 + \chi_Q\left(\frac{k_{\mathrm{rel}}}{k_{\mathrm{HII}}}\right)^{\alpha}\right], \tag{4}$$

where $k_{\mathrm{HII}}$ characterizes the typical size of ionized bubbles.

This modification produces a smoother reionization topology with larger ionized regions and fewer small-scale neutral pockets.

# 5 21-cm Signatures and Observational Probes

The 21-cm line of neutral hydrogen provides a direct probe of the EoR. The brightness temperature contrast is given by

$$\delta T_b \propto x_{\mathrm{HI}}\left(1 - \frac{T_\gamma}{T_s}\right), \tag{5}$$

where $x_{\mathrm{HI}}$ is the neutral fraction and $T_s$ is the spin temperature.

In RDCC, the modified reionization topology and the altered formation history of early galaxies imprint distinctive signatures on the 21-cm power spectrum. The suppression of small-scale modes reduces small-scale fluctuations, while the enhanced coherence of large-scale modes amplifies large-scale structure.

These signatures provide a direct observational window into the relaxation scale and the temporal structure of the gravitational field.

# 6 Conclusion

The Epoch of Reionization in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the formation history of early galaxies, the propagation of ionization fronts, and the topology of reionization. These effects produce distinctive signatures in the ionized volume fraction, the morphology of ionized bubbles, and the 21-cm power spectrum. The present Companion establishes the theoretical foundation for future studies of early galaxy formation, cosmic dawn, and the interaction between radiation and matter in RDCC.

# References


[1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: `https://zenodo.org/records/19744958` (v43.0 or later).

[2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.

[3] Companion XVI: Boltzmann Code and Modified Hierarchy.

[4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.

[5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.

[6] Planck Collaboration (2020), Astron. Astrophys. 641, A6.

[7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$/BAO-Paper].

[8] Maiolino et al. (2024), Nature (JWST high-$z$ galaxies/quasars).

[9] Kaiser, N. (1987), Mon. Not. R. Astron. Soc. 227, 1 (RSD).

[10] Bartelmann, M. & Schneider, P. (2001), Phys. Rep. 340, 291 (Weak Lensing Review).

[11] Hu, W. & Okamoto, T. (2002), Astrophys. J. 574, 566 (CMB Lensing).

[12] Lewis, A. & Challinor, A. (2006), Phys. Rep. 429, 1 (CMB Lensing Review).

[13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), Astrophys. J. 490, 493 (Halo Profiles).